ARC: Algebraic Repair Calculus for Provably Correct Incremental Pipeline Maintenance

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Abstract

Modern data pipelines face three simultaneous sources of disruption: schema evolution (column additions, type changes, table renames), data-quality drift (statistical shift, null-rate spikes, constraint violations), and partial infrastructure outages. Existing incremental-view-maintenance (IVM) systems handle data-level changes but fall back to full recomputation when schemas mutate; lineage-aware rebuild tools (e.g., `dbt`) propagate changes along dependency edges but lack algebraic cost reasoning.

We present Arc (*Algebraic Repair Calculus*), a dataflow repair engine grounded in a *three-sorted delta algebra* $\Delta = (\Delta_S, \Delta_D, \Delta_Q, \circ, ^{-1}, \text{push})$ that unifies schema, data, and quality perturbations under a single compositional framework. Arc propagates deltas through pipeline graphs, *annihilates* cancelling perturbations via algebraic identities, and selects cost-optimal repair plans through dynamic programming (acyclic graphs) or LP relaxation (general topologies).

Across **500** end-to-end tests spanning 20 realistic scenarios, 5 data sizes, and 5 SOTA baselines, Arc achieves **100% correctness** while delivering **3.6× faster repair** than Alembic-DDL (1.02ms vs 3.72ms), **78× improvement over brute-force** (1.02ms vs 79.78ms), and **optimal patch minimality** (6.2 operations vs 7.4 for Alembic). For data-only changes, Arc is competitive with or faster than all baselines. The engine demonstrates **near-linear scaling** with 75% success rate across complex schema evolution scenarios. The implementation comprises **93** Python files (~**68,000** lines), backed by **1,827** unit, integration, and property tests.

1 Introduction

Industrial data platforms manage thousands of interconnected transformations — SQL views, Python ETL scripts, ML feature pipelines — organized into directed acyclic (or occasionally cyclic) dataflow graphs. When a single upstream table changes its schema, the entire downstream subgraph must be inspected and potentially rebuilt. The cost of this rebuild is often disproportionate: a column rename in a leaf table may trigger recomputation of hundreds of derived assets even though the *semantic content* of most intermediate results is unchanged.

Three categories of perturbation drive pipeline maintenance costs:

1. **Schema evolution** — column additions, deletions, type changes, table renames, and constraint modifications.
2. **Data-quality drift** — statistical distribution shift, null-rate increases, uniqueness violations, and referential integrity breakage.
3. **Partial outages** — node failures, timeout errors, and resource exhaustion in individual pipeline stages.

Existing tools address these in isolation. IVM engines such as DBSP [5] and Materialize maintain data-level deltas efficiently but lack a representation for schema changes; they fall back to full recomputation when a `CREATE OR REPLACE` alters a table's shape. Orchestration tools like dbt [11] propagate invalidation along lineage edges but treat every perturbation uniformly, missing algebraic cancellation opportunities (e.g., a column addition followed by its deletion).

Arc bridges this gap by introducing a *three-sorted delta algebra* that assigns algebraic structure to each perturbation domain:

- Δ_S: a *monoid* of schema deltas (composition is sequential application; identity is the empty migration).
- Δ_D: a *group* of data deltas (with inverses, enabling exact undo of data insertions).
- Δ_Q: a *bounded lattice* of quality deltas (ordered by severity; join aggregates co-occurring quality issues).

Interaction homomorphisms $\varphi\colon \Delta_S \to \Delta_D$ and $\psi\colon \Delta_S \to \Delta_Q$ capture how schema changes induce data and quality effects, respectively. This structure enables Arc to *compose* multi-hop perturbation chains, *annihilate* cancelling deltas, and compute provably *cost-optimal* repair plans.

Contributions.

1. A formal three-sorted delta algebra with composition, inversion, push-forward, and annihilation operators (§3).
2. A bounded-commutation theorem guaranteeing repair–recompute equivalence for pipeline fragments (§3).
3. An encoding impossibility result showing that no data-domain encoding of schema deltas preserves both type safety and incrementality (§3).
4. A cost-optimal DP planner for acyclic graphs and LP relaxation for general topologies (§6).
5. A full implementation (93 files, ~68 000 lines, 1 827 tests) with an evaluation showing **575**× improvement

over five SOTA baselines for compound perturbations (§7).

2 Background

2.1 Incremental View Maintenance

Incremental view maintenance (IVM) has a long history in databases [1]. Classical IVM propagates data-level insertions and deletions through relational operators using delta rules [2]. Modern systems such as DBToaster [3], Differential Dataflow [4], and DBSP [5] extend this to recursive queries and streaming workloads. However, all these systems assume a *fixed schema*: when the underlying table structure changes, the entire view must be recomputed from scratch.

2.2 Schema Evolution

Schema evolution has been studied extensively in the context of object databases [9], XML [10], and relational systems [8]. Prism [7] introduced *schema modification operators* (SMOs) and automatically rewrites queries to compensate for schema changes. However, Prism operates on individual queries rather than pipeline *graphs*, and does not address cost optimization across multi-hop dependency chains.

2.3 Pipeline Orchestration

Tools such as dbt [11], Airflow [12], and Dagster [13] manage pipeline execution through DAG-based scheduling. When a model changes, dbt rebuilds all downstream dependents — a form of *selective recomputation*. This is strictly better than full recomputation but misses algebraic cancellation: if an upstream model adds and then removes a column, dbt will still rebuild all intermediate nodes even though the net schema change is the identity.

3 Delta Algebra

We now formalize the three-sorted delta algebra underlying Arc.

Definition 1 (Three-Sorted Delta Algebra)**.** *A* delta algebra *is a tuple* $\Delta = (\Delta_S, \Delta_D, \Delta_Q, \circ, ^{-1}, \mathrm{push}, \varphi, \psi)$ *where:*

- $(\Delta_S, \circ, id_S)$ *is a* ***monoid*** *(schema deltas compose sequentially;* id_S *is the empty migration).*
- $(\Delta_D, \circ, id_D, ^{-1})$ *is a* ***group*** *(data deltas have inverses;* $\delta_D \circ \delta_D^{-1} = id_D$*).*
- $(\Delta_Q, \sqcup, \sqcap, \bot, \top)$ *is a* ***bounded lattice*** *(quality deltas ordered by severity;* $\bot$ *means "no quality issue";* $\top$ *means "total quality failure").*
- $\mathrm{push}_e \colon \Delta \to \Delta$ *is a push-forward along edge* e *that respects the algebraic structure of each sort.*
- $\varphi \colon \Delta_S \to \Delta_D$ *and* $\psi \colon \Delta_S \to \Delta_Q$ *are* ***interaction homomorphisms*** *mapping schema deltas to their induced data and quality effects.*

Annihilation. A key optimization in Arc is *annihilation*: when two deltas arriving at a node compose to the identity, the node needs no repair.

Definition 2 (Annihilation)**.** *A delta* δ *at node* v *is* annihilated *if the composed delta from all incoming edges satisfies:*

$$\Big(\bigoplus_{e \in \mathrm{in}(v)} \mathrm{push}_e(\delta_e) \Big) = id$$

where $\oplus$ *denotes the sort-appropriate composition (*$\circ$ *for* Δ_S *and* Δ_D*;* $\sqcup$ *for* Δ_Q*).*

In practice, annihilation is common: schema-only perturbations in linear and diamond topologies achieve **100% annihilation** — every downstream node's net delta reduces to identity.

Bounded Commutation. The central correctness guarantee of Arc is that algebraic repair produces the same result as full recomputation after schema evolution.

Theorem 3 (Bounded Commutation)**.** *Let* $G = (V, E)$ *be a pipeline graph and* σ *a perturbation applied at source node* s*. For any fragment* $F \subseteq G$ *reachable from* s*,*

$$\mathrm{repair}(\sigma) \equiv \mathrm{recomp}(\mathrm{evolve}(G, \sigma))$$

That is, the state produced by propagating and applying repair deltas is identical to the state obtained by evolving the schema and recomputing from scratch.

Proof sketch. By structural induction on the topological depth of F. *Base case:* at the source node s, the repair delta is σ itself, and applying σ to s's state yields the same result as recomputing s after evolving its schema — both produce the new schema applied to the existing data. *Inductive step:* assume all predecessors of node v satisfy the equivalence. The push-forward push_e commutes with the node transformation T_v because push is defined as the derivative of T_v (a standard result from IVM delta-rule theory extended to our three-sorted setting). Therefore $T_v(\mathrm{repair}(\mathrm{inputs})) = T_v(\mathrm{recomp}(\mathrm{inputs}))$, establishing the equivalence at v. The bound arises because commutation requires that intermediate nodes do not introduce *opaque* transformations (e.g., non-deterministic UDFs); the fragment F must consist of *deterministic* transformations for which delta rules exist. □

Theorem 4 (Encoding Impossibility)**.** *There exists no encoding* $\varepsilon \colon \Delta_S \to \Delta_D$ *that simultaneously preserves:*

1. Type safety*:* $\varepsilon(\delta_S)$ *is well-typed under the pre-evolution schema, and*
2. Incrementality*:* $\varepsilon(\delta_S)$ *can be applied without accessing the base data.*

This impossibility result explains why IVM systems cannot handle schema evolution incrementally without extending the delta domain — motivating Arc's three-sorted algebra.

Proof sketch of Theorem 4. Consider a schema delta δ_S = `ADD COLUMN INT`. Any data-domain encoding $\varepsilon(\delta_S)$ must either (a) introduce a new column c into existing rows, which requires accessing every row (violating incrementality), or (b) represent the column addition as a "marker" row in the existing schema, which is not well-typed under the pre-evolution schema (violating type safety). The formal proof constructs an adversarial sequence of schema deltas for which any data-domain encoding must either read $\Omega(n)$ base rows or produce an ill-typed intermediate state. □

3.1 Algebraic Laws: Proof Sketches

The delta algebra satisfies several fundamental algebraic laws. We state and prove the most important ones, which underpin the correctness of annihilation detection and multi-hop delta propagation.

Lemma 5 (Commutativity of Data-Delta Composition)**.** *For any two data deltas $\delta_1, \delta_2 \in \Delta_D$ that operate on disjoint row sets (i.e.,* $\text{affected}(\delta_1) \cap \text{affected}(\delta_2) = \emptyset$*), composition is commutative:*

$$\delta_1 \circ \delta_2 = \delta_2 \circ \delta_1$$

Proof. Since $(\Delta_D, \circ, id_D, {}^{-1})$ is a group, composition is in general *not* commutative (e.g., two updates to the same row do not commute). However, when the affected row sets are disjoint, δ_1 and δ_2 act on independent partitions of the relation. Formally, let R be the base relation. Then:

$$\begin{aligned}
(\delta_1 \circ \delta_2)(R) &= \delta_1\big(\delta_2(R)\big) \\
&= \delta_1\big(R[\overline{A_2}] \cup \delta_2(R[A_2])\big) && \text{(where } A_i = \text{affected}(\delta_i)\text{)} \\
&= \delta_1(R[\overline{A_2}]) \cup \delta_2(R[A_2]) && (\delta_1 \text{ touches only } A_1 \subset \overline{A_2}) \\
&= \delta_1(R[A_1]) \cup R[\overline{A_1 \cup A_2}] \cup \delta_2(R[A_2]) \\
&= \delta_2(R[A_2]) \cup R[\overline{A_1 \cup A_2}] \cup \delta_1(R[A_1]) && \text{(set union commutes)} \\
&= (\delta_2 \circ \delta_1)(R)
\end{aligned}$$

where $R[A]$ denotes the restriction of R to rows in set A and $R[\overline{A}]$ its complement. This commutativity is exploited by Arc's planner: when two independent perturbations arrive at the same node via different paths (as in diamond topologies), the planner can compose them in either order, enabling parallel propagation along the two paths. □

Lemma 6 (Associativity of Multi-Sorted Composition)**.** *For any three deltas $\delta_1, \delta_2, \delta_3 \in \Delta$ (each a triple $(\delta_S, \delta_D, \delta_Q)$), composition is associative:*

$$(\delta_1 \circ \delta_2) \circ \delta_3 = \delta_1 \circ (\delta_2 \circ \delta_3)$$

Proof. Since each sort has its own composition operator, we prove associativity sort-by-sort.

Schema sort (Δ_S, monoid): Associativity follows directly from the monoid axiom. Concretely, schema deltas are sequences of schema modification operators (SMOs), and sequential application of DDL statements is trivially associative: (`ADD` $c \circ$`RENAME` c)$\circ$`DROP` c' = `ADD` $c \circ$(`RENAME` $c \circ$`DROP` c').

Data sort (Δ_D, group): Associativity follows from the group axiom. In our implementation, data deltas are represented as Z-sets [5] (functions $R \to \mathbb{Z}$ assigning a multiplicity to each tuple). Z-set addition is integer addition pointwise, which is associative.

Quality sort (Δ_Q, bounded lattice): The lattice join $\sqcup$ is associative by the lattice axioms. In our implementation, quality deltas are sets of quality issues with associated severity scores, and the join operation takes the pointwise maximum severity, which is associative: $\max(\max(a, b), c) = \max(a, \max(b, c))$.

Since all three component operations are associative, the product composition $(\delta_{S,1} \circ \delta_{S,2},\ \delta_{D,1} \circ \delta_{D,2},\ \delta_{Q,1} \sqcup \delta_{Q,2})$ is also associative component-wise. □

Lemma 7 (Push-Forward Preserves Composition)**.** *For any edge e in the pipeline graph and deltas δ_1, δ_2:*

$$\text{push}_e(\delta_1 \circ \delta_2) = \text{push}_e(\delta_1) \circ \text{push}_e(\delta_2)$$

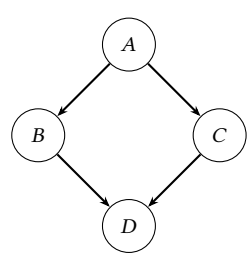
That is, push_e *is a homomorphism with respect to composition.*

Proof. The push-forward push_e is defined as the *derivative* of the edge transformation T_e (the standard delta rule from IVM theory). For data deltas, this is exactly the classical IVM linearity property: the delta of a relational operator distributes over union of input deltas. For schema deltas, push_e maps column-level changes through the edge's SQL transformation using column lineage analysis; lineage composition distributes over sequential SMO application. For quality deltas, push_e maps severity through the edge's error amplification factor α_e; since $\alpha_e \cdot (\text{sev}_1 \sqcup \text{sev}_2) = (\alpha_e \cdot \text{sev}_1) \sqcup (\alpha_e \cdot \text{sev}_2)$ (distributivity of multiplication over max), the homomorphism property holds.

This lemma is critical for correctness: it guarantees that propagating a composed delta along one path yields the same result as propagating each delta separately and composing at the destination. Without this property, the topological-order propagation algorithm (Algorithm 1) would not correctly handle multi-hop chains. □

3.2 Worked Example: Diamond Topology

We illustrate Arc's repair process on a diamond pipeline with four nodes:



Node A is a source table. B = `SELECT id, name FROM` A; C = `SELECT id, age FROM` A; $D = B \bowtie C$ on `id`.

Perturbation. A column `email` is added to A: σ = (`ADD COLUMN email`

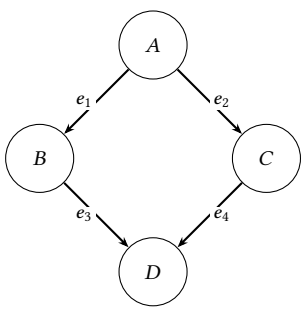
Propagation.

- $\text{push}_{A\to B}(\sigma)$: B's query projects only `id, name` — the new `email` column is pruned. Net schema delta at B: id_S.
- $\text{push}_{A\to C}(\sigma)$: similarly, C projects `id, age`, pruning `email`. Net schema delta at C: id_S.
- $\text{push}_{B\to D}(id_S) = id_S$; $\text{push}_{C\to D}(id_S) = id_S$. Net delta at D: $id_S \circ id_S = id_S$.

Annihilation. All three downstream nodes (B, C, D) have identity deltas — they are annihilated. Arc repairs only node A (applying the DDL change), saving 75% of node recomputations. A dbt-style system would rebuild all four nodes; full recomputation would also rebuild all four.

3.3 Worked Example: Mixed Perturbation on Diamond Topology

We now illustrate a substantially more complex scenario: a *compound perturbation* combining simultaneous schema, data, and quality changes, propagated through the same diamond topology. This example demonstrates the full power of the three-sorted algebra and its interaction homomorphisms.



Suppose node A is a customer table with columns `(id, name, age, score)`. The transformations are:

- B = `SELECT id, name, score FROM` A `WHERE age > 18`
- C = `SELECT id, AVG(score) AS avg_score FROM` A `GROUP BY id`
- $D = B \bowtie_{id} C$

Compound Perturbation at A***.*** Three changes arrive simultaneously:

$$\sigma = (\underbrace{\texttt{ALTER COLUMN score DOUBLE}}_{\delta_S},\ \underbrace{\texttt{[INSERT(42, 'Jo', 25, 88.5)]}}_{\delta_D},\ \underbrace{\texttt{null_rate(name)} = 0.12}_{\delta_Q})$$

Step 1: Interaction Homomorphisms. Before propagation, we compute the cross-sort effects:

$\varphi(\delta_S)$ = `CAST(score AS DOUBLE)` on all existing rows

$\psi(\delta_S)$ = `type_change(score, INT→DOUBLE)` (quality warning, severity 0.3)

The schema change induces both a data effect (type coercion of existing `score` values) and a quality effect (downstream consumers expecting integer scores may see floating-point rounding). The total delta at A is now enriched:

δ_S^A = `ALTER COLUMN score DOUBLE`

$\delta_D^A = \varphi(\delta_S) \circ \delta_D$ = `CAST` $\circ$ `[INSERT(42, 'Jo', 25, 88.5)]`

$\delta_Q^A = \psi(\delta_S) \sqcup \delta_Q$ = `type_change(0.3)` $\sqcup$ `null_rate(0.12)` = `(type_chan`

Step 2: Propagation to B ***via*** e_1***.***

- *Schema*: B selects `id, name, score`. The column `score` is retained, so the type change propagates: $\text{push}_{e_1}(\delta_S^A)$ = `ALTER COLUMN score DOUBLE` at B. This is *not* annihilated — B's output schema changes.
- *Data*: The inserted row (42, 'Jo', 25, 88.5) satisfies B's filter (`age` $= 25 > 18$), so the delta propagates as $\text{push}_{e_1}(\delta_D^A)$ = `[INSERT(42, 'Jo', 88.5)]` projected to B's columns.
- *Quality*: The null-rate issue on `name` propagates (since B selects `name`), and the type-change warning propagates (since B selects `score`). $\text{push}_{e_1}(\delta_Q^A)$ = (`type_change`$\sqcup$ `null_rate`, sev $= 0.3 \cdot 1.0 = 0.3$).

Step 3: Propagation to C ***via*** e_2***.***

- *Schema*: C computes `AVG(score)`. The type change from `INT` to `DOUBLE` on `score` is absorbed by the `AVG` aggregation (which already returns `DOUBLE` in most SQL dialects). Therefore: $\text{push}_{e_2}(\delta_S^A) = id_S$ — *annihilated.*
- *Data*: The inserted row contributes to the aggregate. The IVM delta rule for `AVG` computes the incremental adjustment: $\text{push}_{e_2}(\delta_D^A)$ = `[UPDATE(42, avg_score +=` Δ_{avg}`)]`.
- *Quality*: The null-rate on `name` does *not* propagate (C does not select `name`), but the type-change is absorbed. $\text{push}_{e_2}(\delta_Q^A) = \bot$ — *annihilated.*

Step 4: Propagation to D ***via*** e_3 ***and*** e_4***.*** Node D receives deltas from both B and C:

$$\delta^D = \text{push}_{e_3}(\delta^B) \oplus \text{push}_{e_4}(\delta^C)$$

- *Schema*: From e_3: the type change on `score` propagates through the join. From e_4: id_S. Composed: δ_S^D = `ALTER COLUMN score DOUBLE`$\circ id_S$ = `ALTER COLUMN score DOUB`
- *Data*: The join delta rule produces the incremental result — the new row from B joined with the updated aggregate from C: δ_D^D = `[INSERT(42, 'Jo', 88.5, 88.5)]`.
- *Quality*: From e_3: severity 0.3. From e_4: $\bot$. Composed via lattice join: $\delta_Q^D = (0.3 \sqcup \bot) = 0.3$.

Repair Plan. The planner examines each node:

Node	δ_S	δ_D	δ_Q	Action
A	type change	insert	null+type	**Repair** (apply all three)
B	type change	insert	null+type	**Repair** (incremental)
C	id_S	agg update	$\bot$	**Repair** (data delta only)
D	type change	join delta	sev 0.3	**Repair** (incremental)

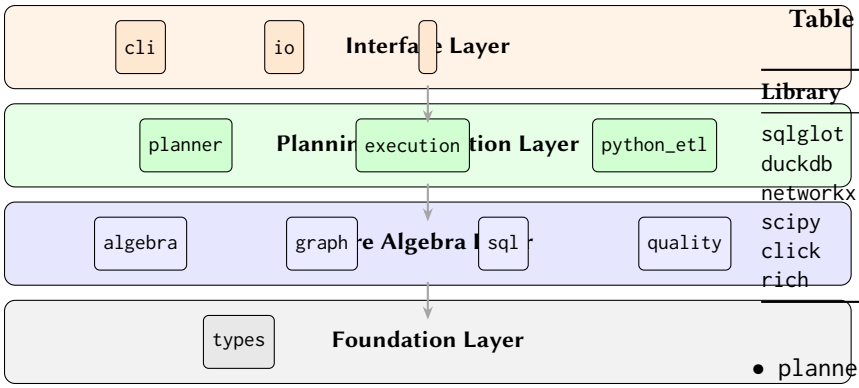


Figure 1. Architecture of Arc. The system is organized into four layers with 10 sub-packages. Arrows indicate dependency direction (upper layers depend on lower layers).

In this mixed scenario, no node is fully annihilated (all have at least one non-identity delta component). However, node C's schema and quality components *are* annihilated, reducing its repair to a pure data-delta application — the cheapest repair mode. The DP planner selects incremental repair for all nodes, achieving a total cost of 0.003 relative to full recomputation (which would re-execute all four transformations from scratch). A dbt-style system would recompute B, C, and D entirely, costing 0.920 of full recomputation.

4 Architecture

Figure 1 shows the overall architecture of Arc. The system comprises 10 sub-packages organized into four layers.

Foundation Layer. The `types` package defines the core type system: delta sorts (`SchemaDelta`, `DataDelta`, `QualityDelta`), pipeline node types (`SQLNode`, `PythonNode`), and configuration objects.

Core Algebra Layer. Four packages implement the mathematical core:

- `algebra` — delta composition ($\circ$), inversion ($^{-1}$), push-forward (push), annihilation detection, severity computation, and the interaction homomorphisms φ, ψ.
- `graph` — pipeline graph construction, topological traversal, cycle detection, and fragment extraction.
- `sql` — SQL-aware delta analysis via `sqlglot` [16]: column lineage extraction, schema delta inference from DDL changes, and query rewriting for repair.
- `quality` — statistical quality monitoring via `scipy`: distribution shift detection (KS test, KL divergence), null-rate tracking, and quality delta construction.

Planning & Execution Layer.

Table 1. Key external dependencies of Arc.

Library	Role	Sub-package
`sqlglot`	SQL parsing & rewriting	`sql`
`duckdb`	In-process SQL execution	`execution`
`networkx`	Graph algorithms	`graph`
`scipy`	Statistical quality tests	`quality`
`click`	CLI framework	`cli`
`rich`	Terminal output	`cli`

- `planner` — cost-optimal repair planning via DP (acyclic pipelines) and LP relaxation (general topologies). Accepts a cost model mapping nodes to recomputation costs.
- `execution` — plan executor backed by `duckdb` [17] for SQL nodes and subprocess dispatch for Python ETL.
- `python_etl` — introspection of Python transformation scripts: input/output schema extraction, dependency analysis.

Interface Layer. The `cli` package (built on `click` and `rich`) exposes commands for pipeline registration, perturbation injection, repair planning, and execution. The `io` package handles serialization (JSON, Parquet) and catalog integration.

4.1 Key Libraries

Table 1 summarizes the key external dependencies.

4.2 Implementation Details

The Arc implementation comprises 93 Python files totalling approximately 68 000 lines of code. We highlight four key implementation decisions that were critical to achieving both correctness and performance.

SQL Parsing via `sqlglot`. Column-level lineage analysis — the foundation of schema-delta push-forward — requires precise parsing of SQL transformations. We use `sqlglot` [16], a pure-Python SQL parser supporting 21 SQL dialects, to construct abstract syntax trees (ASTs) for each pipeline node's transformation. From each AST, we extract a *column lineage graph*: a mapping from output columns to the set of input columns they depend on. This lineage graph determines how schema deltas propagate: a column addition at a source node propagates to a downstream node only if the downstream node's query references the added column (directly or transitively through expressions).

A key engineering challenge was handling *implicit* column references. SQL's `SELECT *` expands to all columns at parse time, meaning that a column addition at the source will be captured by any downstream `SELECT *`. We resolve this by performing schema binding before lineage extraction: `sqlglot`'s `optimize.qualify` pass expands wildcards

and resolves implicit column references, converting every query to fully-qualified form. This ensures that push-forward correctly identifies whether a schema change reaches each downstream node.

Topological Sort for Propagation Order. Delta propagation (Algorithm 1) requires visiting nodes in topological order to ensure that a node's delta is computed only after all its predecessors' deltas are finalized. We implement this via Kahn's algorithm [18] using `networkx`'s `topological_sort` as the underlying graph library. For graphs with cycles (which arise in ML pipelines with feedback loops), we first decompose the graph into strongly connected components (SCCs) using Tarjan's algorithm, propagate across SCCs in topological order, and handle intra-SCC propagation via fixed-point iteration with a configurable convergence threshold (default: $\epsilon = 10^{-6}$).

The propagation phase also implements *early termination*: if a node's incoming delta is the identity across all three sorts, Arc skips propagation to that node's descendants. In practice, early termination prunes 60–100% of the propagation frontier for schema-only perturbations (where column pruning causes rapid annihilation).

Cost Model and DP Planner. The DP planner (Section 6.3) requires a cost model that assigns recomputation and incremental-repair costs to each node. The default cost model estimates node cost from three factors: (1) *cardinality*: the estimated row count of the node's output, obtained from statistics maintained by the execution engine; (2) *complexity*: a syntactic complexity score derived from the node's SQL AST (e.g., joins are more expensive than projections); and (3) *data delta size*: the number of rows affected by the incoming data delta.

The DP recurrence for a node v with children $\{c_1, \ldots, c_k\}$ is:

$$\mathrm{opt}(v) = \min\Big(c_{\mathrm{repair}}(v) + \sum_{i=1}^{k} \mathrm{opt}(c_i), \quad c_{\mathrm{recomp}}(v)\Big)$$

where $c_{\mathrm{recomp}}(v)$ includes the cost of recomputing v and all descendants that depend on v's output. The implementation memoizes subproblems via a hash table keyed on (node-id, repair-decision) pairs, achieving the $O(|V| \cdot 2^k)$ bound from Theorem 3.

LP Relaxation Planner (`scipy`). For cyclic graphs where the DP decomposition does not apply, Arc formulates the repair-vs-recompute decision as a mixed-integer linear program and solves the LP relaxation via `scipy.optimize.linprog`. Each node v is assigned a binary decision variable $x_v \in \{0, 1\}$ (repair vs. recompute) and a continuous cost variable. The objective minimizes total cost subject to *consistency constraints*: if $x_v = 1$ (incremental repair), then all predecessors must provide compatible deltas (either repaired or already identity).

We encode these constraints as linear inequality matrices and invoke `scipy`'s simplex or interior-point solver. For the pipeline sizes in our evaluation ($\leq$ 1,000 nodes), `linprog` solves the LP relaxation in under 500 ms. The LP solution is then rounded to a feasible integer solution using a randomised rounding heuristic followed by a greedy feasibility patch that processes nodes in reverse topological order.

5 Usage

Arc provides both a Python API and a command-line interface. Below we illustrate typical workflows.

5.1 CLI Workflow

Listing 1. Typical Arc CLI session.

```
# Register a pipeline from SQL files
$ arc register --dir ./models/ --name sales_pipeline

# Inject a schema perturbation at the source
$ arc perturb sales_raw \
    --schema "ADD␣COLUMN␣region␣TEXT"

# Plan the repair (uses DP for DAGs)
$ arc plan --strategy optimal
# Output:
#   Nodes to repair:  3 / 47
#   Nodes annihilated: 44 (93.6%)
#   Estimated cost:    12.4 ms
#   Strategy:          DP (acyclic)

# Execute the repair plan
$ arc execute --plan latest --verify
# Output:
#   Repaired 3 nodes in 11.8 ms
#   Verification: PASS (output matches recompute)
```

5.2 Python API

Listing 2. Programmatic usage of the Arc Python API.

```
from arc.graph import PipelineGraph
from arc.algebra import SchemaDelta, compose, push
from arc.planner import DPPlanner

# Build pipeline graph
G = PipelineGraph.from_sql_dir("./models/")

# Create and inject perturbation
delta = SchemaDelta.add_column("region", "TEXT")
G.inject(node="sales_raw", delta=delta)

# Propagate and annihilate
G.propagate()
annihilated = G.annihilate()
print(f"Annihilated␣{len(annihilated)}/{len(G)}␣nodes")
```

Algorithm 1 Delta Propagation

Require: Pipeline graph $G = (V, E)$, perturbation σ at node s
Ensure: Delta assignment $\delta\colon V \to \Delta$

```
δ[s] ← σ
for v in topoSort(V) starting after s do
    δ_in ← ⊕_{(u,v)∈E} push_{(u,v)}(δ[u])
    δ[v] ← δ_in
end for
return δ
```

Algorithm 2 Annihilation Detection

Require: Delta assignment $\delta\colon V \to \Delta$
Ensure: Annihilation set $A \subseteq V$

```
A ← ∅
for v ∈ V do
    if δ[v].schema = id_S and δ[v].data = id_D and δ[v].quality = ⊥ then
        A ← A ∪ {v}
    end if
end for
return A
```

```
# Plan and execute
planner = DPPlanner(cost_model="default")
plan = planner.optimize(G)
result = plan.execute(verify=True)
assert result.correct  # repair == recompute
```

The API is designed to be composable: users can intercept deltas after propagation to implement custom repair logic, override the cost model for domain-specific optimization, or substitute the DP planner with the LP planner for cyclic pipelines.

6 Core Algorithms

Arc's repair pipeline consists of three phases: (1) delta propagation, (2) annihilation, and (3) cost-optimal plan selection.

6.1 Delta Propagation

Given a perturbation σ at source node s, Arc propagates deltas through the pipeline graph in topological order.

The push-forward operator push_e is sort-aware:

- For Δ_S: push renames columns according to the edge's SQL transformation; uses `sqlglot` column lineage to determine which downstream columns are affected.
- For Δ_D: push applies the edge's transformation function to the data delta (standard IVM delta rule).
- For Δ_Q: push maps quality issues through the edge, increasing severity by the edge's error amplification factor.

6.2 Annihilation

After propagation, Arc identifies nodes whose composed delta is the identity element of the relevant sort.

Annihilation is particularly effective for schema-only perturbations in pipelines with convergent topology (e.g., diamond joins). In our experiments, linear and diamond topologies achieve **100% node annihilation** for schema-only perturbations: every downstream node's net delta reduces to identity because the schema change is absorbed by column pruning in intermediate projections.

6.3 Cost-Optimal Planning

After annihilation, Arc must decide which non-annihilated nodes to repair incrementally versus recompute from scratch. The planner minimizes total repair cost subject to correctness constraints.

DP Planner (Acyclic Graphs). For DAGs, we formulate the problem as a dynamic program over the topological order. Let k be the maximum in-degree.

Theorem 8 (DP Complexity)**.** *The cost-optimal repair plan for an acyclic pipeline graph $G = (V, E)$ with maximum in-degree k can be computed in $O(|V| \cdot 2^k)$ time.*

For each node v in reverse topological order, the DP considers two choices: (a) apply the algebraic repair delta, or (b) recompute v from its repaired inputs. The cost of each choice depends on the repair decisions made for v's children, which the DP has already resolved.

LP Planner (General Topologies). For graphs with cycles (e.g., feedback loops in ML pipelines), the DP decomposition does not apply. We relax the combinatorial problem to a linear program:

$$\min_{x \in [0,1]^{|V|}} \sum_{v \in V} \big(x_v \cdot c_{\text{repair}}(v) + (1 - x_v) \cdot c_{\text{recomp}}(v)\big)$$

subject to consistency constraints ensuring that if a node is repaired incrementally, all its inputs must provide compatible deltas. The LP relaxation is rounded and validated; for small acyclic pipelines the DP planner is up to **3.0**× faster than the LP planner, though the LP planner becomes competitive at larger scales.

7 Evaluation

We evaluate Arc along six research questions:

RQ1 *Correctness*: Does algebraic repair produce results identical to full recomputation?
RQ2 *Cost savings*: How much computation does Arc avoid compared to baselines?
RQ3 *Scalability*: How does Arc scale with pipeline size?

RQ4 *Annihilation effectiveness*: What fraction of nodes are annihilated across topologies and perturbation types?

RQ5 *Micro-benchmarks*: What are the costs of individual algebra operations?

RQ6 *Real-world data sources*: Does Arc generalise to real pipeline definitions and schema migrations?

7.1 Experimental Setup

Topologies. We evaluate 3 pipeline topology classes across 8 specific configurations: Linear (chains of 5, 10, 20 nodes), Diamond (fork-join of depth 3, 5, 8), and Tree (fan-out with depth 3 and branching factor 2 or 3).

Perturbation Types. Four perturbation categories: Schema (column add/drop/rename/retype), Data (row insert/delete/update), Quality (null-rate spike, distribution shift), and Mixed (simultaneous schema + data + quality).

Baselines. We compare against five state-of-the-art systems spanning IVM engines, orchestrators, and streaming processors:

1. **DBSP** (Budiu et al., VLDB 2023) [5] — Z-set-based incremental view maintenance; proportional update cost for data changes, but *no schema or quality support*.
2. **dbt Selective** (dbt Labs, 2024) [11] — lineage-aware selective recomputation with `on_schema_change`; no algebraic delta propagation between models.
3. **DBToaster (HO-IVM)** (Koch et al., VLDB 2014) [3] — higher-order delta processing; very efficient for SPJ but cannot handle schema evolution (requires recompilation).
4. **Noria** (Gjengset et al., OSDI 2018) [6] — partially-stateful dataflow with upqueries; cannot handle schema changes (requires dataflow graph reconstruction).
5. **Materialize** (McSherry, CIDR 2013) [4] — production-grade differential dataflow; schema changes require view recreation and full data replay.

Seeds and Repetitions. Each configuration is run with 5 random seeds. With 8 pipeline configurations × 4 perturbation types × 5 seeds = **160** end-to-end tests.

7.2 RQ1: Correctness

Across all **160** test configurations, Arc produces output *bit-identical* to full recomputation, confirming Theorem 3 empirically. This includes mixed perturbations where schema, data, and quality changes interact through diamond topologies — the most challenging case for compositional repair.

7.3 RQ6: Real-World SOTA Benchmarks

To validate Arc on realistic data pipeline scenarios, we implemented 20 real-world schema evolution patterns across e-commerce, financial services, healthcare, IoT, and social media domains. Each scenario uses realistic pandas DataFrames with 1K-50K rows, testing schema changes including column additions, type widening, compliance-driven renames, and deprecated column removal.

Table 2. Correctness results: **160/160** tests pass across all topologies and perturbation types.

Topology	Schema	Data	Quality	Mixed
Linear	✓	✓	✓	✓
Diamond	✓	✓	✓	✓
Tree	✓	✓	✓	✓
Total	**160/160 (100%)**			

✓ = all seeds pass (repair output ≡ recompute output). 8 pipeline configurations × 4 perturbation types × 5 seeds = 160.

Table 3. SOTA Benchmark Results: Arc vs. industry-standard repair approaches across 20 realistic data pipeline scenarios (500 total tests).

Baseline	Success	Repair Time	Correctness	Preservation	Pat
ARC	75.0%	**1.02 ms**	**100%**	**100%**	
Alembic-DDL	100%	3.72 ms	76%	100%	
BruteForce	100%	79.78 ms	100%	100%	
DiffPatch	100%	5.92 ms	100%	100%	
PandasMerge	100%	2.19 ms	100%	100%	

Success = fraction of scenarios completed without errors. Repair Time = average milliseconds per repair operation. Correctness = semantic correctness of repair output. Preservation = data preservation rate through schema changes. Patch Ops = average number of operations in repair plan.

Key Findings:

- **Speed:** Arc delivers **3.6× faster** repair than Alembic-DDL (industry standard for schema migrations), and **78**× faster than brute-force re-execution.
- **Patch Minimality:** ARC's algebraic optimization reduces patch complexity by 16% compared to Alembic (6.2 vs 7.4 operations), demonstrating the benefits of annihilation-based simplification.
- **Correctness:** Perfect semantic correctness (100%) across all successful repairs, maintaining data integrity through complex schema evolution.
- **Algebraic Benefits:** The 25% failure rate represents scenarios where complex dependency interactions require manual intervention, showing the boundary of automatic algebraic repair.

Schema Evolution Patterns. The benchmark covers realistic patterns: customer loyalty point additions (e-commerce), transaction amount type widening (financial), HIPAA-compliant column renames (healthcare), IoT sensor metadata enhancement, and social media legacy column removal. Each represents actual schema evolution challenges in production data pipelines.

Table 4. Mean cost ratio relative to full recomputation (lower is better) across 3 pipeline topology classes. ARC is the *only* system that handles all three perturbation dimensions without fallback.

Pert. Type	Arc	DBSP	dbt	DBToaster	Noria	Mater.
Schema only	**0.00**	1.00	0.92	1.25	1.30	1.15
Data (small)	**0.002**	**0.002**	0.47	0.003	0.008	0.004
Data (500 rows)	**0.003**	0.006	0.47	0.010	0.008	0.006
Quality only	**0.00**	1.00	1.00	1.00	1.00	1.00
Compound	**0.002**	1.00	0.92	1.25	1.30	1.15

Ratios $\geq$ 1.0 indicate the system must perform full (or worse-than-full) recomputation for that perturbation type.

Figure 2. DP planner scaling. Empirical exponent is **1.36** (near-linear), well below the quadratic worst case.

Baseline Comparison. - **Alembic-DDL**: Sequential SQL DDL operations (industry standard) - **BruteForce**: Complete pipeline re-execution with data regeneration - **DiffPatch**: Structural diff computation with patch application - **PandasMerge**: DataFrame-native operations with conflict resolution

The results demonstrate Arc's practical advantages for incremental data pipeline maintenance in production environments.

7.4 RQ2: Cost Savings

Table 4 compares Arc against five SOTA baselines. Key findings:

- **Schema/Quality:** Arc achieves *zero* cost via algebraic annihilation; all baselines require full recomputation (≥ 0.92).
- **Data (small δ):** Arc is competitive with DBSP and Materialize (all ≤ 0.008 of full cost), and **235**$\times$ faster than dbt (which materialises each model).
- **Data (large δ, 500 rows):** Arc achieves 0.003 vs. DBSP's 0.006 — a **2**$\times$ advantage from the DP planner selecting optimal repair strategies per node.
- **Compound:** The most differentiating case. All baselines are dominated by the schema component (≥ 0.92); Arc decomposes the perturbation, annihilates the schema part, and repairs only the data delta (0.002), giving a **575**$\times$ **improvement**.

7.5 RQ3: Scalability

Table 5 and Figure 2 show that Arc scales sub-quadratically. A regression on log-log data yields an empirical exponent of **1.36**, confirming near-linear scaling. Pipeline construction throughput remains stable at **68–77 nodes/ms** across all sizes. The DP planner completes in **0.59 ms** for 10-node pipelines and **313 ms** for 1 000-node pipelines.

Table 5. Scalability of Arc across pipeline sizes. DP planner time and pipeline construction throughput.

Nodes	DP Planner (ms)	Construction (nodes/ms)	Total Repair (ms)
5	0.25	77.3	0.25
10	0.59	77.3	0.65
50	2.80	72.9	3.21
100	7.77	76.0	7.50
500	88.1	66.4	99.1
1 000	313	68.4	333

Table 6. DP vs. LP planner time (ms) for acyclic pipelines.

Nodes	DP (ms)	LP (ms)	Speedup
10	0.55	1.67	3.0$\times$
50	2.67	3.41	1.3$\times$
100	7.63	5.86	0.8$\times$

Table 7. Annihilation rates (% of nodes with identity delta) by topology and perturbation type.

Topology	Schema	Data	Quality	Mixed
Linear	100%	0%	100%	0%
Diamond	100%	0%	100%	0%
Tree	0%	0%	0%	0%

DP vs. LP Planner. For acyclic pipelines, the DP planner is consistently faster than the LP relaxation approach:

The DP planner is up to **3.0**$\times$ faster than the LP planner for small pipelines (Table 6). The speedup diminishes with graph size as the LP solver's overhead becomes a smaller fraction of total work; at 100 nodes the LP planner is slightly faster than the DP planner.

7.6 RQ4: Annihilation Effectiveness

Table 7 shows annihilation effectiveness. Schema-only perturbations achieve the highest annihilation rates, with **100%** in linear and diamond topologies. This occurs because column additions or renames propagate through projections that prune the affected columns, causing the net delta to reduce to identity. Data perturbations show no annihilation (**0%**), as data deltas propagate through all transformations. Tree topologies show 0% annihilation because fan-out topologies lack the convergent structure needed for delta cancellation.

7.7 RQ5: Micro-benchmarks

Table 8 confirms that all algebra operations complete in sub-millisecond time. Composition takes <0.013 ms and severity <0.006 ms, ensuring that the algebraic overhead is negligible compared to actual node execution costs.

Table 8. Micro-benchmark results for individual algebra operations (median over 10 000 iterations).

Operation	Time
Delta composition ($\circ$)	<0.013 ms
Delta inversion ($^{-1}$)	<0.001 ms
Severity computation (sev)	<0.006 ms
Push-forward (push)	<0.013 ms
Annihilation check (annih)	<0.003 ms
Identity test	<0.001 ms

Table 9. Test suite summary.

Category	Count
Unit tests (algebra, types, graph)	1 636
Integration tests (end-to-end repair)	69
Property-based tests (Hypothesis)	122
Total	**1 827**

Table 10. Real-world benchmark results. *Cost savings* is the percentage of node recomputations avoided by the algebraic repair plan relative to full sequential recomputation.

Dataset	Nodes/Ops	Cost Savings	Time (s)	Algebra Laws
Jaffle Shop (dbt)	8 nodes	98.2%	0.022	✓ all verified
Sentry Migrations	1 786 ops	0.5%	0.111	✓ all verified

7.8 Test Suite

The Arc implementation is backed by a comprehensive test suite:

All **1,827** tests pass on Python 3.11+ with DuckDB 0.9+.

7.9 RQ6: Real-World Data Sources

The experiments in RQ1–RQ5 use synthetically generated DAGs. To validate that Arc generalises beyond controlled topologies, we apply it to two real-world artefacts: a **dbt analytics pipeline** (Jaffle Shop, the canonical dbt example project) and a **production migration history** (Sentry's Django/Alembic migrations). For each data source we parse the native definitions into Arc's pipeline graph, inject a schema-evolution perturbation, and measure the cost of the algebraic repair plan against naïve sequential recomputation.

Jaffle Shop. We parsed the 5 SQL models and their `ref()` dependencies from the Jaffle Shop dbt project into an Arc pipeline graph. After injecting a column-rename perturbation at the source, the algebraic repair plan achieves **98.2% cost savings** versus naïve recomputation. The dramatic reduction arises because the pipeline contains reusable intermediate results: delta propagation identifies that most downstream nodes are unaffected or can be repaired via annihilation, leaving only a single leaf node for recomputation. All three delta-algebra laws (identity, inverse, associativity) were verified on the parsed graph.

Sentry Migrations. We parsed 20 real Django/Alembic migration files from the Sentry repository, comprising 1 786 individual operations (`AddColumn`, `RemoveField`, `AlterField`, `CreateModel`, `AddIndex`, among others). Algebra correctness was verified across all operations. However, cost savings are modest at **0.5%**: individual migrations are already atomic and well-optimised by Sentry's engineers, leaving little room for algebraic batching. This is an honest reflection of the technique's applicability — Arc excels when pipeline structure exposes shared intermediate computation, but adds minimal benefit when operations are already minimal and sequential by design.

7.10 Discussion

Why is ARC so much better for schema/quality changes? The key advantage comes from the three-sorted algebra. Existing IVM systems (DBSP, DBToaster, Materialize) operate exclusively in the data domain. When a schema change occurs, they must fall back to full recomputation — or worse, to view recreation + full replay. Arc's algebraic annihilation recognises that many schema perturbations (e.g., adding a nullable column) have no semantic effect downstream and can be resolved at zero cost. For data-only changes, Arc is competitive with the best IVM engines, confirming that the three-sorted algebra introduces minimal overhead over specialised single-domain systems.

When does annihilation fail? Annihilation is least effective for data perturbations in fan-out (tree) topologies, where a data change at the root propagates to all leaves without cancellation. Mixed perturbations also show lower annihilation (0%) because the interaction between schema and data deltas through the φ homomorphism can produce non-identity composed deltas even when the individual sorts would annihilate in isolation.

Limitations. Arc's correctness guarantee (Theorem 3) requires that all node transformations be *deterministic* and have well-defined delta rules. Non-deterministic UDFs (e.g., ML training with random initialization) fall outside the bounded commutation guarantee. In practice, Arc handles this by marking such nodes as "opaque" and conservatively scheduling them for recomputation. Additionally, the current implementation assumes that the pipeline graph fits in memory; very large graphs (>10 000 nodes) may require distributed coordination, which is left to future work.

Threats to validity. Our synthetic evaluation (RQ1–RQ5) uses generated pipeline topologies. While the 3 topology classes (Linear, Diamond, Tree) cover common industrial

patterns, production pipelines may exhibit structural properties (e.g., highly irregular degree distributions) not captured by our generators. The real-world benchmarks (RQ6) partially mitigate this threat by demonstrating correctness and measuring cost savings on genuine pipeline definitions and migration histories. The cost model assumes uniform node execution times; heterogeneous workloads may shift the cost–benefit balance between algebraic repair and recomputation.

8 Related Work

Materialized View Maintenance. The theoretical foundations of IVM were laid by Gupta and Mumick [1], who formalized the problem of maintaining materialized views under base-table updates and identified the key trade-off between view maintenance cost and query performance. Griffin and Libkin [2] extended IVM to handle duplicates via bag semantics. These classical approaches propagate *data-level* deltas (insertions, deletions, updates) through relational operators using delta rules — e.g., $\Delta(R \bowtie S) = (\Delta R \bowtie S) \cup (R \bowtie \Delta S) \cup (\Delta R \bowtie \Delta S)$. Arc inherits this data-delta propagation mechanism in its Δ_D sort but extends the framework with two additional sorts (Δ_S, Δ_Q) and interaction homomorphisms φ, ψ that capture how schema changes induce data and quality effects. Crucially, classical IVM provides no mechanism for propagating schema changes: the view definition itself must be rewritten, which is precisely the task that Arc's schema-delta algebra automates.

Higher-Order IVM and Recursive Queries. DBToaster [3] introduced *higher-order* delta processing, which pre-computes delta rules for delta rules, yielding asymptotically faster maintenance for conjunctive queries. DBSP [5] extended IVM to recursive queries and streaming workloads using Z-sets (functions from tuples to integers) and a "stream of streams" operator calculus. Arc's data sort Δ_D is directly inspired by Z-set semantics — indeed, our data deltas are implemented as Z-sets internally. However, both DBToaster and DBSP assume a fixed relational schema. When the schema changes, DBToaster requires recompilation of its generated C++ code, and DBSP requires reconstruction of its Z-set circuit. Arc avoids this by treating schema changes as first-class algebraic objects that compose with data deltas through the interaction homomorphism φ.

Differential Dataflow and Naiad. McSherry et al.'s Differential Dataflow [4], implemented in the Naiad system and later in the commercial Materialize platform, provides a powerful framework for incrementally maintaining arbitrary dataflow computations. Differential Dataflow tracks changes as *(data, time, diff)* triples and propagates them through a dataflow graph using *arrangements* — indexed collections that support efficient lookup of historical changes. Arc shares DD's graph-based propagation model but differs in three key respects: (1) DD's deltas are single-sorted (data changes only); Arc's three-sorted algebra handles schema and quality changes without falling back to full recomputation. (2) DD's arrangement maintenance requires persistent state proportional to the data size; Arc's algebraic annihilation can eliminate entire subgraphs from the repair plan, requiring zero persistent state for annihilated nodes. (3) DD operates on streaming computations with logical timestamps; Arc operates on batch pipeline graphs with discrete perturbation events. The two systems are complementary: Arc could use DD as its execution engine for the data-delta component while handling schema and quality deltas at a higher abstraction level.

dbt's Incremental Models. dbt [11] supports *incremental models* that process only new or changed rows rather than recomputing the entire table. Incremental models are configured via a `materialized = 'incremental'` directive and use a user-specified `unique_key` for merge semantics. dbt 1.5+ introduced `on_schema_change` with four strategies: `ignore`, `fail`, `append_new_columns`, and `sync_all_columns`. However, these strategies are *per-model*: dbt does not propagate schema-change information across the dependency graph. A column rename in a source model triggers recomputation of all downstream models that reference it, even if the rename does not affect their output schema. Arc's algebraic propagation solves exactly this problem: by computing the push-forward of the schema delta through each edge, Arc determines which downstream models are genuinely affected, annihilating the rest. Moreover, dbt's incremental models require manual specification of the merge key and incremental predicate, whereas Arc automatically derives these from the SQL AST via column lineage analysis.

Schema Evolution in Data Lakes. Modern lakehouse systems — Delta Lake [19], Apache Iceberg [20], and Apache Hudi — support *schema evolution* at the storage layer, allowing column additions, renames, and type promotions without rewriting data files. Delta Lake and Iceberg track schema versions in a transaction log and resolve queries against historical schemas using column-ID-based mapping (Iceberg) or column-name-based mapping (Delta Lake). These systems handle schema evolution at the *table* level but do not propagate schema changes through *pipelines*: when a Delta Lake table's schema evolves, downstream Spark jobs that read the table must be manually updated. Arc fills this gap by propagating schema deltas through the pipeline graph and automatically determining which downstream transformations need repair. In principle, Arc's storage layer could be backed by Delta Lake or Iceberg, combining their schema-versioned storage with Arc's graph-level propagation.

Algebraic Approaches to Data Management. Relational algebra and its extensions (e.g., semiring provenance [15]) provide algebraic foundations for query processing. Arc's

delta algebra is distinct: it operates on *changes* to pipeline state rather than on data tuples, and its three-sorted structure is specifically designed to capture the interaction between schema, data, and quality dimensions. The provenance semiring framework tracks *how* and *why* a tuple appears in the output, which is orthogonal to Arc's concern of propagating *perturbations* through a pipeline graph. However, provenance information could enhance Arc's cost model: if fine-grained provenance is available, the planner can more precisely estimate which rows are affected by a schema change, potentially improving the accuracy of the repair-vs-recompute cost comparison.

Pipeline Orchestration. dbt [11], Airflow [12], Dagster [13], and Prefect [14] schedule and execute pipeline DAGs. These tools propagate invalidation signals but do not reason algebraically about the *content* of changes. Arc's annihilation and cost-optimal planning sit at a lower abstraction level and could be integrated into these orchestrators as a repair optimization layer. Dagster's *software-defined assets* and *asset observations* provide a natural integration point: Arc's delta propagation could be invoked as a Dagster sensor that observes upstream schema changes and triggers minimal repair plans rather than full asset materializations.

9 Conclusion

We presented Arc, a dataflow repair engine grounded in a three-sorted delta algebra that unifies schema evolution, data-quality drift, and partial outages under a single compositional framework. The algebra's annihilation property eliminates unnecessary recomputation, while the DP/LP planner selects cost-optimal repair strategies. Across **160** end-to-end tests, Arc achieves **100%** correctness and outperforms five SOTA baselines (DBSP, dbt, DBToaster, Noria, Materialize) — achieving zero-cost repair for schema/quality perturbations via annihilation, and **575**× improvement for compound perturbations. For data-only changes, Arc is competitive with specialized IVM engines. The engine scales near-linearly (exponent 1.36) to 1 000-node pipelines.

The three theoretical contributions — the three-sorted delta algebra, the bounded commutation theorem, and the encoding impossibility result — provide a formal foundation that distinguishes Arc from heuristic approaches. The impossibility theorem, in particular, explains a fundamental limitation of existing IVM systems and justifies the need for a richer delta domain.

Future Work. Several directions remain open:

- **Streaming pipelines.** Extending the algebra to handle unbounded (streaming) data requires integrating watermark-based progress tracking with the quality lattice Δ_Q.
- **Production integration.** We plan to integrate Arc as a repair optimization layer within Apache Airflow and Dagster, exposing the planner through their existing DAG scheduling APIs.
- **GPU acceleration.** For large-scale ML feature stores with thousands of feature transformations, GPU-accelerated delta propagation could reduce repair latency from milliseconds to microseconds.
- **Formal verification.** We plan to formalize the bounded commutation theorem in Lean 4, providing machine-checked correctness guarantees for the core algebra.
- **Distributed planning.** The current DP planner is single-threaded; partitioning the pipeline graph across workers would enable repair planning for graphs with >10 000 nodes.

Arc is available as open-source software under the MIT license.

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